7. 다음과 같은 이산 주기 신호의 이산 시간 푸리에 급수를 구하라.

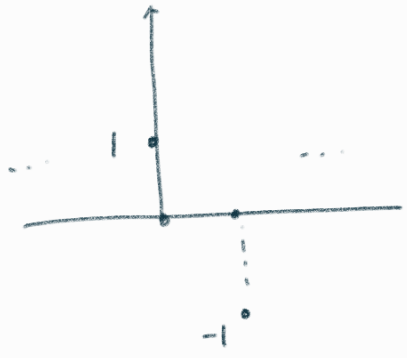
(a) $x[n]=\begin{cases}1, & n=\text{짝수}\\ 0, & n=\text{홀수}\end{cases}$

(b) $x[n]=\begin{cases}+1, & n=\text{짝수}\\ -1, & n=\text{홀수}\end{cases}$

(c) $x[n]=\sum_{m=-\infty}^{\infty}\delta[n-mN]$

(d) $x[n]=\sum_{m=-\infty}^{\infty}(\delta[n+1-mN]+\delta[n-mN]+\delta[n-1-mN])$

(b)



$x[n] = [1, -1]$ 의 푸리에 변환

$N = 2$

$$X_0 = \frac{1}{2}\sum_{n=0}^{1} x[n]$$

$$= \frac{1}{2}(1-1) = 0$$

$$X_1 = \frac{1}{2}\sum_{n=0}^{1} x[n]e^{j\Omega_0 n}$$

$$= \frac{1}{2}(1-1\cdot e^{j\Omega_0})$$

$$= \frac{1}{2}(1-e^{j\Omega_0})$$

$$\Omega_0 = \frac{2\pi}{2} = \pi$$

$$\therefore X_1 = \frac{1}{2}(1-e^{j\pi})$$

$$= \frac{1}{2}\cdot e^{\frac{j\pi}{2}}\cdot(e^{-\frac{j\pi}{2}} - e^{\frac{j\pi}{2}})$$

$$= \frac{1}{2}\cdot j\cdot(e^{-\frac{j\pi}{2}} - e^{\frac{j\pi}{2}})$$

$$= \frac{1}{2}(e^{\frac{j\pi}{2}} - e^{-\frac{j\pi}{2}})$$

$$= \frac{1}{2j}\left(e^{j\frac{\pi}{2}n} - e^{-j\frac{\pi}{2}n}\right)$$

$$= \sin\left(\frac{\pi}{2}n\right) = 1$$

$$\therefore\ x[n] = e^{j\pi n}$$

(d)

$$X_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$$

$$= \frac{1}{N}\left(e^{-j\frac{2\pi}{N}k(N-1)} + 1 + e^{-j\frac{2\pi}{N}k}\right)$$

$$= \frac{1}{N}\left(e^{-j(2\pi k - \frac{2\pi}{N}k)} + 1 + e^{-j\frac{2\pi}{N}k}\right)$$

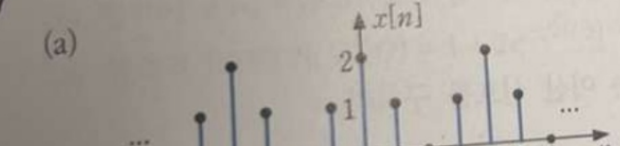
$$= \frac{1}{N}\left(e^{j\frac{2\pi}{N}k} + 1 + e^{-j\frac{2\pi}{N}k}\right)$$

$$= \frac{1}{N}\left(1 + 2\cos\left(\frac{2\pi}{N}k\right)\right)$$

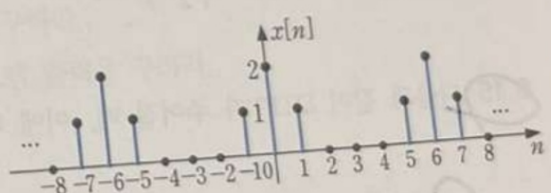
$$\therefore\ x[n] = \sum_{k=0}^{N-1} \frac{1}{N}\left(1+2\cos\left(\frac{2\pi}{N}k\right)\right)e^{jk\frac{2\pi}{N}n}$$

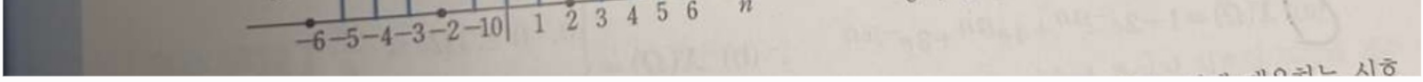
6.9 다음 그림과 같은 이산 주기 신호에 대해서 이산 시간 푸리에 급수의 계수를 구하고, 진폭 및 위상 스펙트럼을 그려라.

(a)



(b)





(a)

$x[n] = [2, 1, 0, 1]$ $\quad N = 4$ $\quad \Omega = \frac{2\pi}{4} = \frac{\pi}{2}$

$$X_k = \frac{1}{4} \sum_{n=0}^{3} x[n] e^{-jk\frac{\pi}{2}n}$$

$$X_0 = \frac{1}{4}(2 + 1 + 0 + 1) = 1$$

$$X_1 = \frac{1}{4}\left(2 + 1 \cdot e^{-j\frac{\pi}{2}} + 0 + 1 \cdot e^{-j\frac{3}{2}\pi}\right)$$

$$= \frac{1}{4}\left(2 + e^{-j\frac{\pi}{2}} + e^{j\frac{\pi}{2}}\right)$$

$$= \frac{1}{2} + \frac{1}{2}\cos\left(\frac{\pi}{2}\right) = \frac{1}{2}$$

$$X_2 = \frac{1}{4}\left(2 + 1 \cdot e^{-j\pi} + 0 + 1 \cdot e^{-j3\pi}\right)$$

$$= \frac{1}{4}(2 - 1 - 1) = 0$$

$$X_3 = \frac{1}{4}\left(2 + 1 \cdot e^{-j\frac{3}{2}\pi} + 0 + 1 \cdot e^{-j\frac{9}{2}\pi}\right)$$

$$= \frac{1}{4}\left(2 + e^{j\frac{\pi}{2}} + 0 + e^{-j\frac{1}{2}\pi}\right)$$

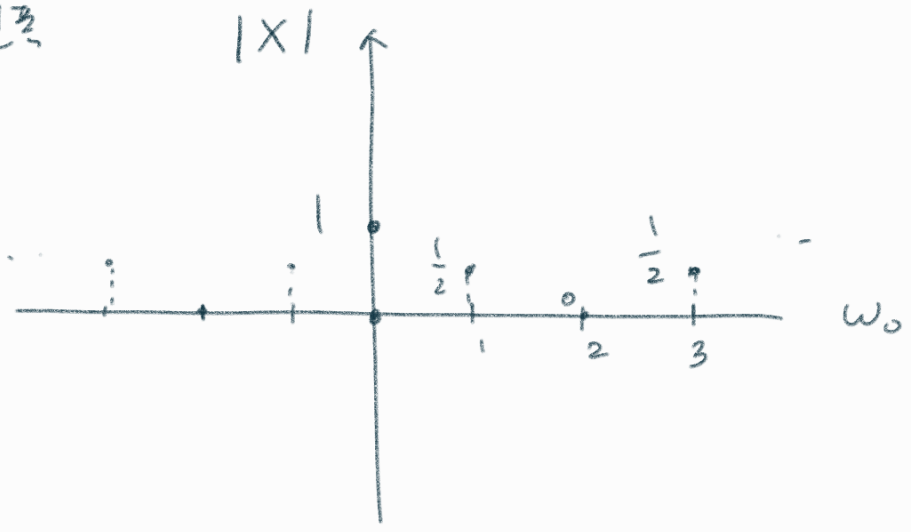
$$= \frac{1}{2} + \frac{1}{2}\cos\left(\frac{\pi}{2}\right) = \frac{1}{2}$$

$X_k = [1, \frac{1}{2}, 0, \frac{1}{2}]$

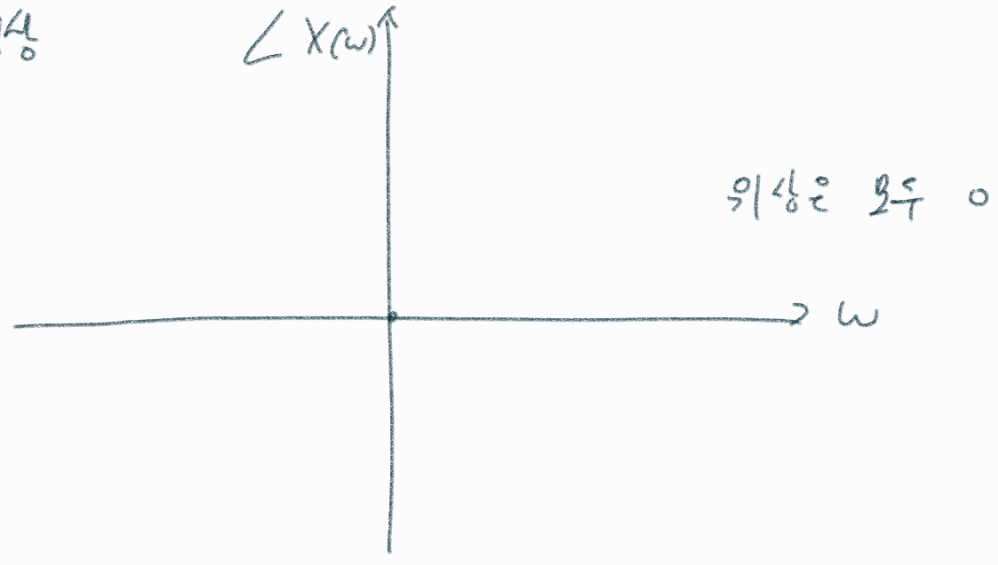
$$\therefore \quad x[n] = \sum_{k=0}^{3} X_k e^{jk\frac{\pi}{2}n}$$

$$= 1 + \frac{1}{2}e^{j\frac{\pi}{2}n} + \frac{1}{2}e^{j\frac{3}{2}\pi n}$$

크기



위상



(b)

$$x[n] = [2, 1, 0, 0, 0, 1]$$

$N = 6 \qquad \Omega = \frac{\pi}{3}$

$$X_k = \frac{1}{6} \sum_{n=0}^{5} x[n] e^{-jk\frac{\pi}{3}n}$$

$$X_0 = \frac{1}{6}(2+1+1) = \frac{4}{6} = \frac{2}{3}$$

$$X_1 = \frac{1}{6}\left(2 + 1\cdot e^{-j\frac{\pi}{3}} + 1\cdot e^{-j\frac{5}{3}\pi}\right)$$

$$= \frac{1}{6}\left(2 + e^{-j\frac{\pi}{3}} + e^{j\frac{\pi}{3}}\right) = \frac{1}{3} + \frac{1}{3}\cos\left(\frac{\pi}{3}\right)$$

$$= \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

$$X_2 = \frac{1}{6}\left(2 + e^{-j\frac{2}{3}\pi} + e^{-j\frac{10}{3}\pi}\right)$$

$$= \frac{1}{3} + \frac{1}{3}\cos\left(\frac{2}{3}\pi\right) = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$$

$$X_3 = \frac{1}{6}\left(2 + e^{-j\pi} + e^{-j5\pi}\right)$$

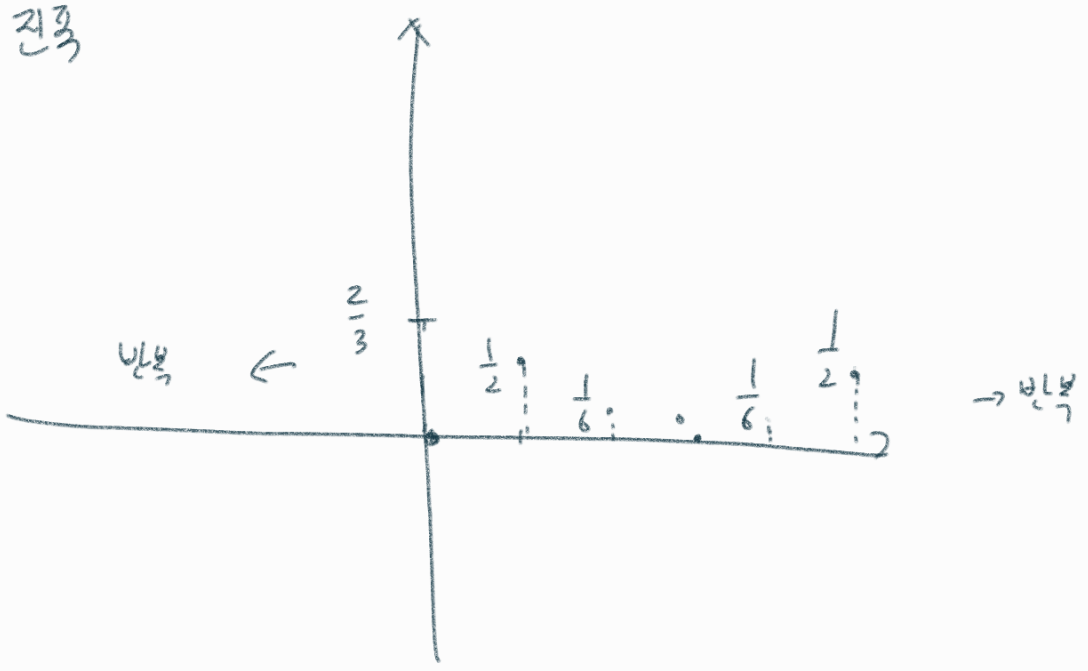
$$= \frac{1}{3} + \frac{1}{3}\cos(\pi) = 0$$

$$X_4 = \frac{1}{6}\left(2 + e^{-j\frac{4}{3}\pi} + e^{-j\frac{20}{3}\pi}\right)$$

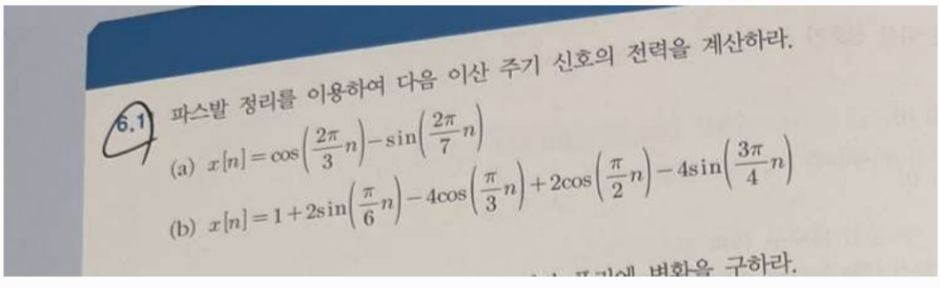
$$= \frac{1}{3} + \frac{1}{3}\cos\left(\frac{4}{3}\pi\right) = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$$

$$X_5 = \frac{1}{2}$$

$$X_k = \left[\frac{2}{3}, \frac{1}{2}, \frac{1}{6}, 0, \frac{1}{6}, \frac{1}{2}\right]$$



위상 스펙트럼은 없음.



6.1 파스발 정리를 이용하여 다음 이산 주기 신호의 전력을 계산하라.

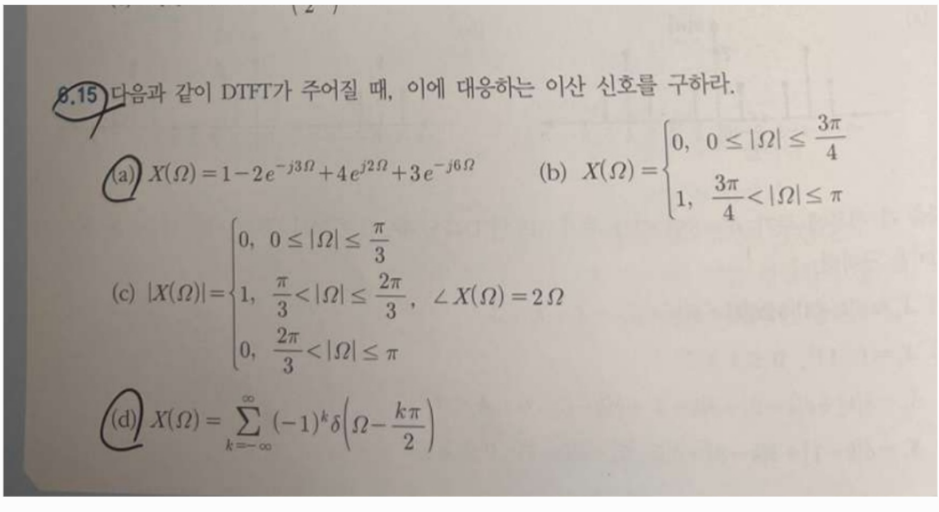
(a) $x[n]=\cos\left(\frac{2\pi}{3}n\right)-\sin\left(\frac{2\pi}{7}n\right)$

(b) $x[n]=1+2\sin\left(\frac{\pi}{6}n\right)-4\cos\left(\frac{\pi}{3}n\right)+2\cos\left(\frac{\pi}{2}n\right)-4\sin\left(\frac{3\pi}{4}n\right)$

에 변화을 구하라.

(a) $\frac{1}{2}+\frac{1}{2}=1$

(b) $1+2+8+2+8=21$



6.15 다음과 같이 DTFT가 주어질 때, 이에 대응하는 이산 신호를 구하라.

(a) $X(\Omega)=1-2e^{-j3\Omega}+4e^{j2\Omega}+3e^{-j6\Omega}$

(b) $X(\Omega)=\begin{cases}0, & 0\le|\Omega|\le\frac{3\pi}{4}\\ 1, & \frac{3\pi}{4}<|\Omega|\le\pi\end{cases}$

(c) $|X(\Omega)|=\begin{cases}0, & 0\le|\Omega|\le\frac{\pi}{3}\\ 1, & \frac{\pi}{3}<|\Omega|\le\frac{2\pi}{3}\\ 0, & \frac{2\pi}{3}<|\Omega|\le\pi\end{cases}$, $\angle X(\Omega)=2\Omega$

(d) $X(\Omega)=\sum_{k=-\infty}^{\infty}(-1)^k\delta\left(\Omega-\frac{k\pi}{2}\right)$

(a) $e^{j\Omega_0 t} \Leftrightarrow 2\pi\delta(\Omega-\Omega_0)$

$2\pi\, e^{-j\Omega_0\Omega} \Leftrightarrow 2\pi\delta(t-\Omega_0)$

$e^{-j\Omega_0\Omega} \Leftrightarrow \delta(n-\Omega_0)$

$x[n] = \delta[n] - 2\delta[n-3] + 4\delta[n+2] + 3\delta[n-6]$

(d) $\frac{e^{j\Omega_0 n}}{2\pi} \Leftrightarrow \delta(\Omega-\Omega_0)$

$$\therefore x[n] = \sum_{k=-\infty}^{\infty} (-1)^k \frac{e^{j\frac{\pi}{2}n}}{2\pi}$$

$$= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} (-1)^k e^{j\frac{\pi}{2}kn}$$

(d) $y[n]-0.3y[n-1]-0.4y[n-2]=2x[n-1]-x[n-2]$

6.18 입력 $x[n]=\delta[n]+0.5\delta[n-1]$에 대한 출력이 $y[n]=\delta[n]-2\delta[n-1]-\delta[n-2]$와 같은 이산 시스템에 대해 다음을 구하라.

(a) 이산 시스템의 주파수 응답

(b) 이산 시스템의 임펄스 응답

(c) 이산 시스템을 나타내는 차분 방정식 표현

(a) $X(\omega) = 1 + \frac{1}{2}e^{-j\omega}$

$Y(\omega) = 1 - 2e^{-j\omega} - e^{-2j\omega}$

$$H(\omega) = \frac{1 - 2e^{-j\omega} - e^{-2j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$$

(6b)

$$H(w)\left(1+\frac{1}{2}e^{-jw}\right) = 1-2e^{-jw}-e^{-2jw}$$

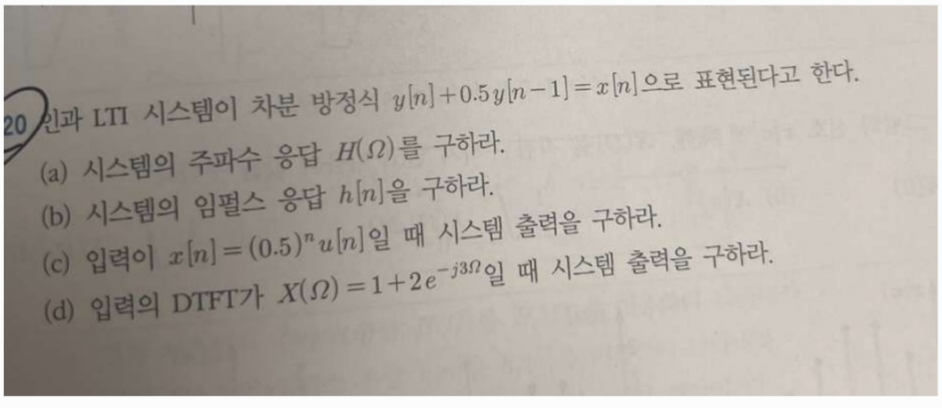
$$h(w)(\delta[n]+\frac{1}{2}\delta[n-1])$$

$$= \delta[n]-2\delta[n-1]-\delta[n-2]$$

$$h[n]+\frac{1}{2}h[n-1] = \delta[n]-2\delta[n-1]-\delta[n-2]$$

(c)

$$y[n]+\frac{1}{2}y[n-1] = x[n]-2x[n-1]-x[n-2]$$



20 인과 LTI 시스템이 차분 방정식 $y[n]+0.5y[n-1]=x[n]$으로 표현된다고 한다.

(a) 시스템의 주파수 응답 $H(\Omega)$를 구하라.

(b) 시스템의 임펄스 응답 $h[n]$을 구하라.

(c) 입력이 $x[n]=(0.5)^n u[n]$일 때 시스템 출력을 구하라.

(d) 입력의 DTFT가 $X(\Omega)=1+2e^{-j3\Omega}$일 때 시스템 출력을 구하라.

(a) $$H(\Omega) = \frac{1}{1+0.5e^{-j\Omega}}$$

$$= \frac{1}{1+\frac{1}{2}e^{-j\Omega}} = \frac{2}{2+e^{-j\Omega}}$$

(b)

$$H(\Omega)\cdot(2+e^{-j\Omega}) = 2$$

$$h[n]+\frac{1}{2}h[n-1] = 2\delta[n]$$

$h(\Omega) \cdot (2 \cdot \delta[n] + \delta[n-1] \quad 2\delta[n]$

$2h[n] + h[n-1] = 2\delta[n]$

$h[n] + \frac{1}{2}h[n-1] = \delta[n]$

$h[0] = 1$

$h[1] + \frac{1}{2} = 0$

$h[1] = -\frac{1}{2}$

$h[2] = \frac{1}{4}$ ...

$\therefore h[n] = \left(-\frac{1}{2}\right)^n u[n] = (-0.5)^n u[n]$

(c) $x[n] = (0.5)^n u[n]$

$$X(\Omega) = \frac{1}{1 - \frac{1}{2}e^{-j\Omega}}$$

$$H(\Omega) = \frac{1}{1 + \frac{1}{2}e^{-j\Omega}}$$

$$X(\Omega) \cdot H(\Omega) = \frac{1}{1 - \frac{1}{4}e^{-j2\Omega}} = Y(\Omega)$$

$$Y(\Omega) = \frac{1}{1 - \frac{1}{2}e^{-j\Omega}} \cdot \frac{1}{1 + \frac{1}{2}e^{-j\Omega}}$$

$y[n] = \frac{1}{2}(-0.5)^n u[n] + \frac{1}{2}(0.5)^n u[n]$

(2)

$x[n] = \delta[n] + 2\delta[n-3]$

$y[n] = h[n] * x[n]$

$= (0.5)^n u[n] + 2(-0.5)^{n-3} u[n-3]$

7.7 다음에 주어진 $X[k]$에 대하여 IDFT를 수행하여 $x[n]$을 구하라.

(a) $X[k] = [1,\ j,\ 0,\ -j]$

(b) $X[k] = [1,\ 1,\ 1,\ 1]$

(c) $X[k] = [1,\ -1,\ 1,\ -1]$

(d) $X[k] = [2,\ 0,\ -2,\ 0]$

(e) $X[k] = [1,\ 0,\ 0,\ j,\ 0,\ -j,\ 0,\ 0]$

(f) $X[k] = \begin{cases} 3, & k = 0 \\ 1, & 1 \le k \le 9 \end{cases}$

(a) $X^*[k] = [1, -j, 0, j]$

$$x[k]^* = \sum_{n=0}^{3} X^*[n] W_4^{kn}$$

$x[0]^* = 1 - j + 0 + j = 1$

$x[1]^* = 1 - 1 - 1 = -1$

$x[2]^* = 1 + j + 0 - j = 1$

$x[3]^* = 1 + 1 + 0 + 1 = 3$

$\therefore\ x[n] = \left[\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$

(b) $X^*[k] = [1, -1, 1, -1]$

$x[0]^* = 1 - 1 + 1 - 1 = 0$

$x[1]^* = 1 + i - 1 - i = 0$

$x[2]^* = 1 + 1 + 1 + 1 = 4$

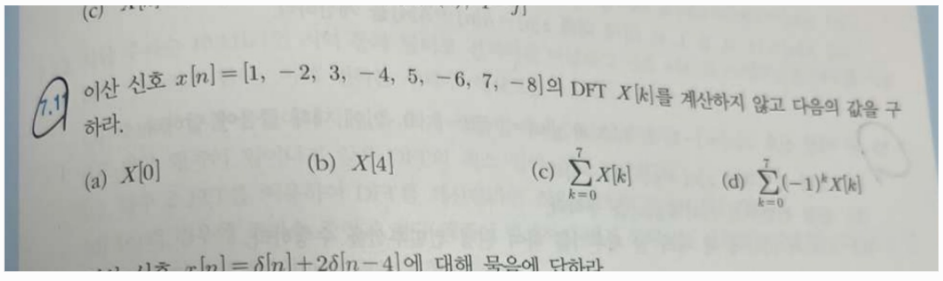
$x[3]^* = 1 - i - 1 + i = 0$

$\therefore x[n] = [0, 0, 1, 0]$

(e) $X^*[k] = [1, 0, 0, -j, 0, j, 0, 0]$

$x[n] = \frac{1}{8}\left(1 + je^{j\frac{3\pi}{4}n} - je^{j\frac{5\pi}{4}n}\right)$

$\therefore x[n] = \left[\frac{1}{8}, \frac{1-\sqrt{2}}{8}, \frac{3}{8}, \frac{1-\sqrt{2}}{8}, \frac{1}{8}, \frac{1+\sqrt{2}}{8}, -\frac{1}{8}, \frac{1+\sqrt{2}}{8}\right]$



(c)

7.1 이산 신호 $x[n] = [1, -2, 3, -4, 5, -6, 7, -8]$의 DFT $X[k]$를 계산하지 않고 다음의 값을 구하라.

(a) $X[0]$ (b) $X[4]$ (c) $\sum_{k=0}^{7} X[k]$ (d) $\sum_{k=0}^{7} (-1)^k X[k]$

... 신호 $x[n] = \delta[n] + 2\delta[n-4]$에 대해 물음에 답하라.

a) $X[0] = \sum_{n=0}^{7} x[n] = 1 - 2 + 3 - 4 + 5 - 6 + 7 - 8$

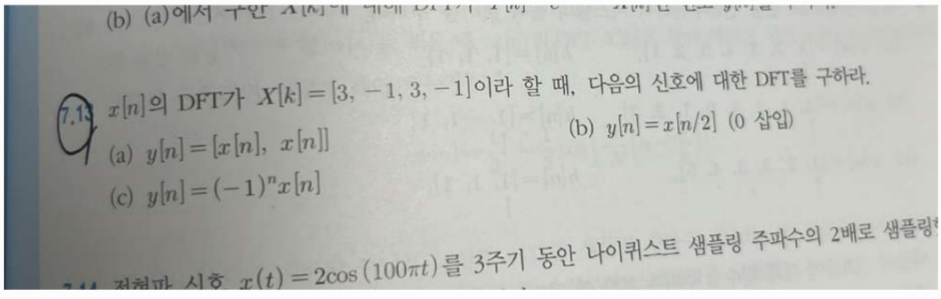
$= -1 - 1 - 1 - 1 = -4$

b) $X[4] = \sum_{n=0}^{7} x[n] \cdot W_8^{4n}$

$$= 1+2+3+4+5+6+7+8$$
$$= 36$$

c) $1 \cdot 8 = 8$

d) $5 \cdot 8 = 40$



(b) (a)에서 구한 $X[k]$에 대해 DFT가 $x[n]$인 신호 $y[n]$

7.13 $x[n]$의 DFT가 $X[k]=[3,-1,3,-1]$이라 할 때, 다음의 신호에 대한 DFT를 구하라.

(a) $y[n]=[x[n],\ x[n]]$

(b) $y[n]=x[n/2]$ (0 삽입)

(c) $y[n]=(-1)^n x[n]$

7.14 전형파 신호 $x(t)=2\cos(100\pi t)$를 3주기 동안 나이퀴스트 샘플링 주파수의 2배로 샘플링

(a) $y[n]=[x[n/2], x[n/2]]$

$$Y_8[k]=\sum_{r=0}^{n-1} e^{-j\frac{2\pi}{n}kr}$$

$$X_4[k \bmod 4]=(1+(-1)^{(k)})\ X_4[k \bmod 4]$$

$$Y_8[k]=\begin{cases} 2X_4[\frac{k}{2}] \\ 0 \end{cases}$$

$$Y_8 = [6, 0, -2, 0, 6, 0, -2, 0]$$

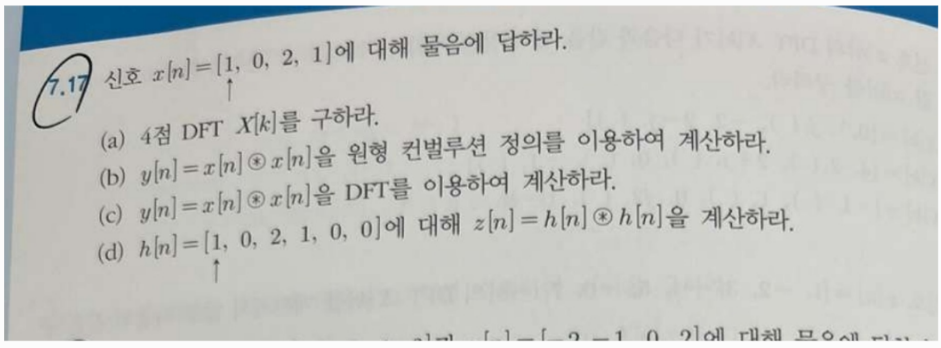
(b) $$Y_8[k]=\sum_{m=0}^{3} x[m]e^{-j\frac{2\pi}{8}k(2m)} = \sum_{m=0}^{3} x[m]e^{-j\frac{2\pi}{4}km}$$

$$= X_4[k \bmod 4]$$

$$Y_8 = [3, -1, 3, -1, 3, -1, 3, -1]$$

(c) $$Y_4[k]=X_4[(k-\tfrac{N}{2}) \bmod 4] = X_4[(k-2) \bmod 4]$$

$y_4 = [3, -1, 3, -1]$



7.17 신호 $x[n] = [1, 0, 2, 1]$에 대해 물음에 답하라.

(a) 4점 DFT $X[k]$를 구하라.

(b) $y[n] = x[n] \circledast x[n]$을 원형 컨벌루션 정의를 이용하여 계산하라.

(c) $y[n] = x[n] \circledast x[n]$을 DFT를 이용하여 계산하라.

(d) $h[n] = [1, 0, 2, 1, 0, 0]$에 대해 $z[n] = h[n] \circledast h[n]$을 계산하라.

$x[n]$

(a) $N = 4$

$$X[K] = [4, -1+i, 2, -1-i]$$

(b) $[1, 0, 2, 1] \circledast [1, 0, 2, 1]$

$$y[n] = [5, 4, 5, 2]$$

(c)

$$X[k] \cdot X[k]$$

$$= [16, -2i, 4, 2i]$$

→ IDFT $[16, 2i, 4, -2i]$

$$y[0] = \frac{1}{4}(20) = 5$$

$$y[1] = \frac{1}{4}(16 + 2 - 4 + 2) = 4$$

$$y[2] = \frac{1}{4}(20) = 5$$

$$y[3] = \frac{1}{4}(16 - 2 - 4 - 2) = 2$$

$$[5, 4, 5, 2]$$

(2) [1, 0, 2, 1, 0, 0]

[1, 0, 2, 1, 0, 0]

z_n = [2, 0, 4, 2, 4, 4]

7.18 두 이산 신호 $x[n]=[-1, 2, 1, 3]$과 $y[n]=[-2, -1, 0, 2]$에 대해 물음에 답하라.

(a) 선형 컨벌루션 $x[n] * y[n]$을 구하라.

(b) 원형 컨벌루션 $x[n] \circledast y[n]$을 구하라.

(c) $x[n]$과 $y[n]$에 세 개의 영 채우기를 하여 원형 컨벌루션을 수행하라.

(a) [-1, 2, 1, 3]

[2, 0, -1, -2]

-2 0 -1 -6

$x[n] * y[n]$ = [2, -3, -4, -9, 1, 2, 6]

(b)

[-1, 2, 1, 3] [-2, -1, 0, 2]

$\circledast$ = [3, -1, 2, -9]

(c)

$x[n]$ = [-1, 2, 1, 3, 0, 0, 0]

$y[n]$ = [-2, -1, 0, 2, 0, 0, 0]

$\circledast$ = [2, -3, -4, -9, 1, 2, 6]

7.19 LTI 시스템의 임펄스 응답 $h[n]$과 입력 $x[n]$이 다음과 같이 주어질 때, 중복-더하기 방법과 중복-

저장 방법으로 선형 컨벌루션하여 시스템의 출력 $y[n]$을 구하라.

(a) $x[n]=[\underset{\uparrow}{1},\ 2,\ 3,\ 4,\ 3,\ 2,\ 1]$, $h[n]=[\underset{\uparrow}{1},\ 1,\ 1]$

(b) $x[n]=[\underset{\uparrow}{1},\ 1,\ 2,\ 3,\ 0,\ 1,\ 2,\ 3]$, $h[n]=[\underset{\uparrow}{1},\ -1,\ 1]$

(c) $x[n]=[\underset{\uparrow}{1},\ 2,\ 3,\ 3,\ 4,\ 5]$, $h[n]=[\underset{\uparrow}{1},\ 1,\ 1]$

(a) $l=2$ $l+m-1=4$

① 중복-더하기

1) [1, 2, 0, 0] [1, 1, 1, 0] = [1, 3, 3, 2]

2) [3, 4, 0, 0] [1, 1, 1, 0] = [3, 7, 7, 4]

3) [3, 2, 0, 0] [1, 1, 1, 0] = [3, 5, 5, 2]

4) [1, 0, 0, 0] [1, 1, 1, 0] = [1, 1, 1, 0]

∴ [1, 3, 6, 9 10, 9, 6, 3, 1, 0]

① 중복-저장

1) [0, 0, 1, 2] → [~~3~~, ~~2~~, 1, 3]

2) [1, 2, 3, 4] → [~~8~~, ~~7~~, 6, 9]

3) [3, 4, 3, 2] → [~~8~~ 9, 10, 9]

4) [3, 2, 1, 0] → [~~4~~ 5, 6, 3]

5) [1, 0, 0, 0] → [1, 1, 1, 0]

[1, 3, 6, 9, 10, 9, 6, 3, 1, 0]

(b) 중복-더하기

1) [1, 1, 0, 0] → [1, 0, 0, 1]

2) [2, 3, 0, 0] → [2, 1, -1, 3]

3) [0, 1, 0, 0] → [0, 1, -1, 1]

4) [2, 3, 0, 0] → [2, 1, -1, 3]

[1, 0, 2, 2, -1, 4, 1, 2, -1, 3]

- 중복 - 리턴

1) [0, 0, 1, 1] → [0, 0, 1, 0]

2) [1, 1, 2, 3] → [1, 0, 2, 2]

3) [2, 3, 0, 1] → [2, 1, -1, 4]

4) [0, 1, 2, 3] → [0, 1, 1, 2]

5) [2, 3, 0, 0] → [2, 1, -1, 3]

[1, 0, 2, 2, -1, 4, 1, 2, -1, 3]

C) 중복 - 더하기

1) [1, 2, 0, 0]

[1, 3, 3, 2]

2)

[3, 6, 6, 5]

3) [4, 5, 0, 0]

[4, 9, 9, 5]

[1, 3, 6, 8, 10, 12, 9, 5]

중복 - 과장

1) [0, 0, 1, 2]

[3, 2, 1, 3]

2) [1, 2, 3, 3]

[1, 1, 1, 0] → [7, 6, 6, 8]

3) [3, 3, 4, 5]

[12, 11, 10, 12]

4 [4, 5, 0 0]

[4, 9, 9, 5]

[1, 3, 6, 8, 10, 12, 9, 5]

7.20 다음의 신호들에 대해 밑수 2 주파수 분할 역 비트순 FFT, 밑수 2 시분할 역 비트순 FFT를 이용하여 샘플 스펙트럼을 각각 구하고, 이에 대한 FFT 선도를 그려라.

(a) $x[n] = [1, 2, 3, 2]$

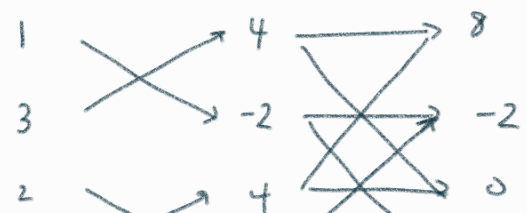
(b) $x[n] = [1, 0, -1, 1, -1, 1, 0, -1]$

(c) $x[n] = [1, -1, 1, -1]$

(d) $x[n] = [1, 2, 3, 2, 1, 0, 0, 0]$

【응용 문제】

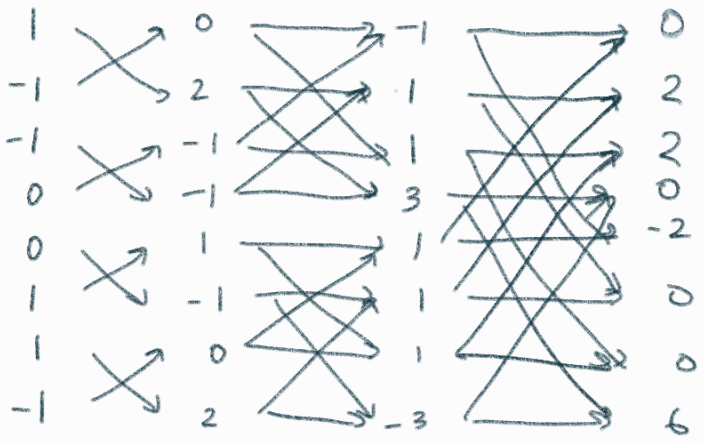
(a) 0, 2, 1, 3





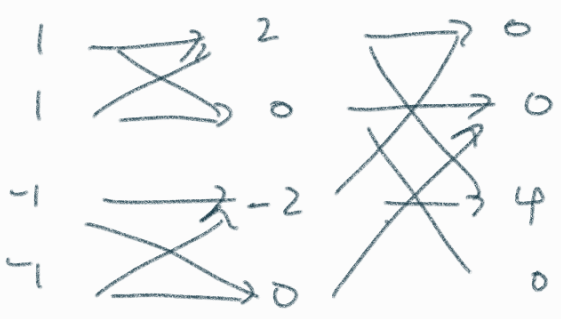
[8, -2, 0, -2]

(b) 0, 4, 2, 6 / 1, 5, 3, 7



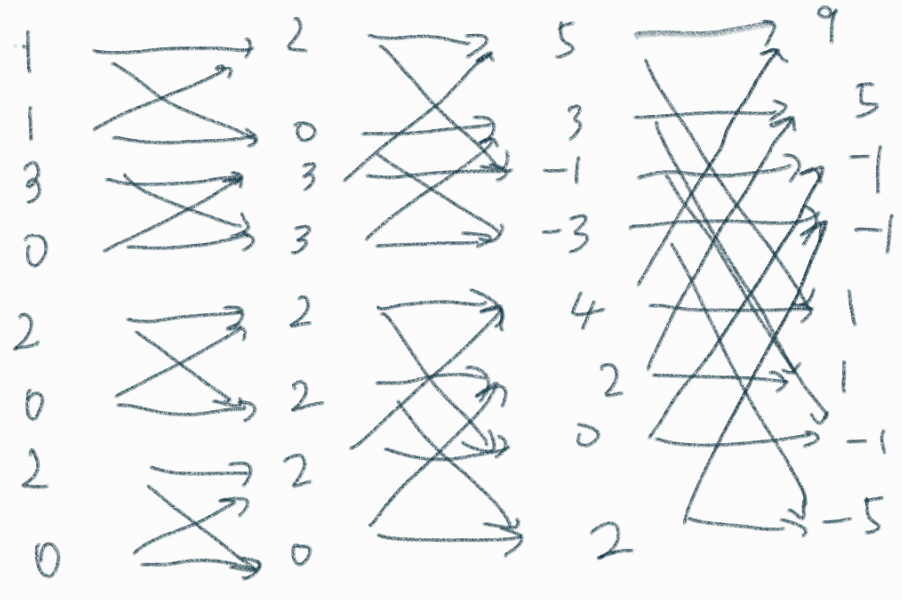
[0, 2, 2, 0, -2, 0, 0, 6]

(c)



[0, 0, 4, 0]

(d)



[9, 5, -1, -1, 1, 1, -1, -5]